

Grounded SAM (v1) — local evaluation on NVIDIA GeForce RTX 3080

Pipeline: **text prompt -> Grounding DINO 1.0 (local) -> box -> SAM v1 (local) -> mask**.
Fully local. No cloud API, no API key, no image upload, no network at inference.

This is the SAM v1 sibling of the Grounded SAM 2 evaluation. It was run so the two can be compared on identical inputs. **The Grounding DINO detector is byte-identical between the two** (same build, config, checkpoint, thresholds, seed): across all 50 RefDrone pairs the predicted boxes matched with **0 mismatches**, so every difference below is attributable to the SAM stage alone.

Bottom line — classification: REFERENCE ONLY for this use case. Grounded SAM (v1) is a strictly worse fit than Grounded SAM 2 here, on every axis that differs, and identical on the one axis that decides the headline task.

- Referring-expression accuracy is **22%, exactly equal to Grounded SAM 2** — because the failure is in the shared detector, not in SAM. Swapping SAM 2 for SAM v1 cannot fix it.
- SAM v1's mask stage is **4.0x slower and uses ~2.5x the VRAM** of SAM 2.1 Tiny, for masks of equivalent visual quality (same box, same image).
- **SAM v1 has no video memory/propagation at all**. Applied to video it must re-detect every frame (4.1 FPS vs SAM 2's 32 FPS) and maintains **no object identity** — the top-1 mask hops between instances (13 cars/frame; 27 hops in 200 frames).

Full reasoning and the head-to-head table in sections 5–8 and 11.

1. What was actually run

Phase	What	Status
1	Smoke test, single aerial image, prompts <code>car / person</code>	PASS
2	RefDrone sample, 50 (image, expression) pairs, seed 1234	PASS (accuracy identical to SAM 2 — see 5)
3	Video, 2 clips, PER-FRAME detect+segment (SAM v1 has no propagation)	PASS (found the no-identity behaviour — see 6)
4	Benchmark, config A (SAM v1 ViT-B) vs B (SAM v1 ViT-H)	PASS

Nothing here is claimed on the basis of a script exiting 0. Every result below was checked against the rendered masks or the per-frame mask geometry.

2. Environment

GPU	NVIDIA GeForce RTX 3080 (sm_86, 11911 MB), driver 580.95.05
CPU / RAM	12th Gen Intel(R) Core(TM) i9-12900K / 62.5 GB
Python / torch	3.12.3 / 2.5.1+cu124 (CUDA 12.4)
transformers	4.44.2
GD CUDA extension	COMPILED
SAM v1 checkpoints	ViT-B (93.7 M), ViT-H (641.1 M)
Cloud SDK installed	no

The detector is the exact build from the Grounded SAM 2 evaluation, imported from `third_party/grounded_sam_2` with its already-compiled `MultiScaleDeformableAttention` CUDA op. Only the segmentation stage (`segment_anything` , from `third_party/grounded_sam`) is new. This is deliberate: holding the detector fixed is what makes the comparison an apples-to-apples measurement of the SAM stage.

3. Local-only verification

- Cloud SDK (`dds_cloudapi_sdk`): **not installed**.
- `assert_no_cloud_imports()` runs after every inference call and raises if any `dds_cloudapi` / `dinox` module is ever present in `sys.modules` . It never fired.
- **No upstream submodule file was modified**. All work lives in a wrapper, `scripts/gsam_local.py` .
- `gsam_local.device()` raises rather than run on CPU, so no CPU timing is reported as GPU.

4. Network posture (inherited from the detector)

Same finding as the SAM 2 evaluation, and for the same reason: Grounding DINO's text encoder calls `from_pretrained("bert-base-uncased")` , and transformers revalidates the cached files against `huggingface.co` on every load. `gsam_local` sets `HF_HUB_OFFLINE` / `TRANSFORMERS_OFFLINE=1` at import, before transformers loads, so the pipeline is network-free by default. The BERT encoder (~421 MB) lives in `~/.cache/huggingface` , not in `checkpoints/` , and must be staged on an air-gapped target. SAM v1 itself adds no new network dependency.

5. Phase 2 — RefDrone referring expressions (identical to SAM 2, and that is the point)

50 pairs, seed 1234, split 30 train / 10 val / 10 test. Config: GD Swin-T + SAM v1 `vit_b`, `box_thr` 0.25, `text_thr` 0.2.

Metric	Grounded SAM (v1)	Grounded SAM 2
Top-1 localization, IoU>=0.5	22.0%	22.0%
Mean best IoU	0.212	0.212
Median best IoU	0.004	0.004
No-detection rate	4.0%	4.0%
Multiple-detection rate	84.0%	84.0%
Mean GD latency	104 ms	104 ms
Mean SAM latency	107 ms (ViT-B, fp32)	26 ms (Tiny, bf16)
Peak VRAM	3518 MB	1433 MB

The localization columns are identical **to four decimal places** because the boxes are identical (0 mismatches over 50 pairs). This is the single most important thing this evaluation shows: **the referring-expression failure is entirely in Grounding DINO 1.0**, which resolves the category and discards the qualifiers. It is a shared, detector-side capability gap. **No choice of SAM version touches it.** The only thing SAM v1 changes is that it draws the (equally wrong-or-right) mask 4.0x slower.

The characteristic failure is unchanged: "The black cars near the intersection." returns 10 boxes, top-1 IoU 0.00, and SAM v1 segments that wrong box perfectly (figure 02). A clean, confident mask on the wrong object — same as SAM 2, because it is the same box.

Metric caveats (same as the SAM 2 run). This is my protocol, not RefDrone's official metric: highest-confidence box, scored as max IoU over the GT boxes. Do not compare to published RefDrone numbers. RefDrone provides boxes only — no mask ground truth — so SAM v1 mask quality is assessed visually (figure 01) and is deliberately **not scored**.

6. Phase 3 — video: SAM v1 cannot track, so it re-detects every frame

There is no SAM 2 protocol to run here, because SAM v1 has **no video predictor**: no `init_state`, no `propagate_in_video`, no memory. The only coherent way to apply Grounded SAM (v1) to a clip is to run the full image pipeline on every frame. Two things fall out.

6a. Cost — the detector is in the loop on every frame

Clip	Throughput	ms/frame (GD + SAM)	vs SAM 2 same clip
tracking_car.mp4 (1920x1080, 200 frames)	4.1 FPS	211 (103 + 108)	31 FPS (propagation)
zebra.mp4 (1280x720, 200 frames)	4.3 FPS	210 (104 + 106)	—

SAM 2 pays Grounding DINO **once** (358 ms) and then every subsequent frame is a cheap propagation step; Grounded SAM (v1) pays GD + SAM on all 200 frames. That is the whole architectural difference, and it is worth ~8x in throughput on this hardware.

6b. No object identity — the top-1 mask hops between instances

Because each frame is independent, "which object" is re-decided every frame by detector confidence. There is no id linking frame N to frame N-1.

	tracking_car.mp4	zebra.mp4
Detection rate	100%	100%
Mean detections / frame	13.0 (max 17)	5.9 (max 7)
Top-1 centroid jump (median / max)	9 / 746 px	4 / 698 px
Frame-to-frame hops (> 10% diagonal)	27	20

Detection rate is 100% only because these clips always contain some instance of the class; it says nothing about tracking a specific one. Figures 04-05 show the mechanism: between frame 192 and 193 the top-1 car mask jumps ~746 px from a car in the centre lanes to a different car near the horizon — a clean mask on a different object.

How this compares to the SAM 2 failure mode. SAM 2's Phase 3 found a silent identity switch: after the tracked car left the frame at f69, SAM 2 re-bound object id 1 to a different car 953 px away and kept reporting it as the same track. That is arguably worse because it is hidden — SAM 2 claims a persistent identity and then quietly breaks it. Grounded SAM (v1) never makes that claim: it has no persistent identity to break, so the instance-hopping is visible in every frame rather than concealed. Neither is usable as a single-target tracker out of the box; they fail differently. SAM 2 fails by lying about an identity it has; SAM v1 fails by never having one.

7. Phase 4 — performance (ViT-B vs ViT-H, NVIDIA GeForce RTX 3080)

3 warm-ups discarded, 10 measured image iterations, 120 per-frame video frames. Image input 1920x1080 (prompt car , 7 boxes); video input 1280x720.

	A: GD Swin-T + SAM v1 ViT-B	B: GD Swin-T + SAM v1 ViT-H
SAM params	94 M	641 M
Model load (cold)	1941 ms	4607 ms
First inference (cold)	459 ms	796 ms
Warm image latency	217 ms (4.6 FPS)	559 ms (1.8 FPS)
of which Grounding DINO	101 ms	102 ms
of which SAM v1	116 ms (enc 108 + dec 8)	457 ms (enc 448 + dec 9)
p95 image latency	218 ms	560 ms
Per-frame video throughput	4.4 FPS (227 ms/frame)	1.8 FPS (570 ms/frame)
Peak VRAM (image)	3518 MB	6541 MB
Peak VRAM (video)	3518 MB	6541 MB
Peak CPU RSS	1879 MB	2018 MB

All GPU regions are timed with `torch.cuda.synchronize()` on **both** sides. Each config ran in its own fresh process so B's cold-load time is not flattered by A's warm CUDA context.

Unlike SAM 2 (where the detector dominates and the SAM backbone barely moves the needle), **the SAM backbone is a real lever here.** ViT-B -> ViT-H nearly triples image latency (217 -> 559 ms) and adds 3023 MB of VRAM, almost entirely in the image encoder (enc 108 -> 448 ms). ViT-H's 6541 MB peak also matters on an 11 GB card. On this evidence there is no reason to prefer ViT-H: no mask-quality gain was observed in any run, and the box it segments is chosen by the (identical) detector regardless.

8. Head-to-head: Grounded SAM (v1) vs Grounded SAM 2

Same detector, same inputs, same machine. Everything below is measured.

	Grounded SAM (v1)	Grounded SAM 2	Winner
RefDrone top-1 @IoU0.5	22%	22%	tie (shared detector)
SAM mask latency (RefDrone image)	107 ms	26 ms	SAM 2
Warm image latency	217 ms (4.6 FPS)	143 ms (7.0 FPS)	SAM 2
Video approach	detect every frame	detect once + propagate	SAM 2
Video throughput	4.1 FPS	32 FPS	SAM 2
Cross-frame object identity	none (per-frame)	memory-based*	SAM 2
Peak VRAM (image, smallest)	3518 MB	1433 MB	SAM 2
Mask quality (given a correct box)	excellent	excellent	tie

*SAM 2's memory is a real capability but it silently switches identity after target egress (section 6b) — better than v1's no-identity, still not a drop-in single-target tracker.

The verdict is not close. On the headline task (referring expressions) the two are **identical** because the weak stage is shared; on every axis where they differ — mask cost, VRAM, video throughput, and the existence of any temporal identity at all — **SAM 2 wins**, most of them by a wide margin.

9. What works (SAM v1)

- Fully local, GPU, no network, no cloud API. No upstream file modified.
- **SAM v1 mask quality is excellent** — tight, clean masks on aerial imagery (figure 01), visually indistinguishable in quality from SAM 2. Given a correct box it is not the weak link. The problem is never mask quality; it is cost and the absence of tracking.
- 100% per-frame detection on clips that always contain the class.

10. What does not work / does not exist (SAM v1)

- **Referring-expression grounding: 22%**, identical to SAM 2 — a shared detector-side capability gap, unfixable by any SAM choice.
- **No video tracking capability at all.** No memory, no propagation, no object id. "Tracking" degrades to per-frame re-detection: 4.1 FPS and instance-hopping.
- **Expensive masks.** 4.0x the SAM 2 latency and ~2.5x the VRAM for no quality gain.
- **ViT-H buys nothing** here but ~2x the latency and 3023 MB more VRAM.

11. Classification: REFERENCE ONLY (for this use case)

Scoped, and stated plainly:

REFERENCE ONLY as a detect-and-track pipeline for this project. Grounded SAM (v1) is a competent image detect-and-segment pipeline and its masks are excellent, but it is dominated by Grounded SAM 2 on this workload. It shares SAM 2's fatal referring-expression weakness (same detector, 22%) while giving up SAM 2's one genuine strength — cheap, memory-based video tracking — because SAM v1 has no video capability to begin with. For a single-image, single-shot segmentation task where 100 ms and 3.5 GB are acceptable, it is perfectly serviceable. For anything involving video, or where VRAM/latency matter, or where the goal is referring-expression grounding, there is no reason to choose it over Grounded SAM 2, and several reasons not to.

The one thing this evaluation establishes cleanly, and that carries beyond the v1-vs-v2 question: **the 22% referring-expression ceiling is the detector's, not the segmenter's.** Replacing Grounding DINO 1.0 is the only lever that matters; both SAM versions are waiting on a better box.

12. Reproduce

```
source .venv-grounded-sam2/bin/activate # same env: torch+CUDA, groundingdino, segment_anything
on path
python experiments/grounded_sam_local/scripts/check_environment.py
python experiments/grounded_sam_local/scripts/run_image_demo.py \
    --image datasets/refdrone/images/all_image/0000001_02999_d_0000005.jpg \
    --prompts "car" "person"
python experiments/grounded_sam_local/scripts/run_refdrone_sample.py --seed 1234
python experiments/grounded_sam_local/scripts/run_video_demo.py \
    --video third_party/grounded_sam_2/assets/tracking_car.mp4 --prompt "car"
python experiments/grounded_sam_local/scripts/run_video_demo.py \
    --video third_party/grounded_sam_2/assets/zebra.mp4 --prompt "zebra" --start-frame 20
python experiments/grounded_sam_local/scripts/benchmark_pipeline.py
python experiments/grounded_sam_local/scripts/make_figures.py
python experiments/grounded_sam_local/scripts/summarize_results.py
```

13. Figures

See [figures/](#). SAM v1 mask quality (01), the shared grounding failure (02), small aerial targets (03), and the per-frame no-identity hop between consecutive frames (04, 05), plus per-frame zebra (06).

14. Not measured / not claimed

- SAM v1 mask **accuracy** is unscored: RefDrone has no mask ground truth and none was invented. Quality is asserted visually only.
- No comparison against published RefDrone baselines (different metric — see section 5).
- Jetson performance: **not measured**.
- Multi-object per-frame association / a real tracker on top of v1: not built, not claimed. Only single top-1 was followed, precisely to demonstrate the absence of identity.

Figure gallery



Figure 01. SAM v1 SEGMENTATION IS GOOD. 'The blue vehicles in the image.' -> 1 box, IoU 0.84 vs GT, and the SAM v1 (ViT-B) mask is tight on the vehicle. Given a correct box, SAM v1 is not the weak link -- same conclusion as SAM 2. The difference is cost, not quality: this mask took ~107 ms vs SAM 2's ~26 ms.



Figure 02. THE FAILURE IS IN THE DETECTOR, AND IT IS SHARED. 'The black cars near the intersection.' -> 10 boxes, top-1 IoU 0.00. Grounding DINO resolved the CATEGORY (cars) and ignored the referring qualifiers, then SAM v1 segmented the wrong box perfectly. This is the SAME detector and the SAME box the SAM 2 run scored (0 box mismatches over all 50 pairs), so this failure is identical in both pipelines -- swapping the SAM version changes nothing here.

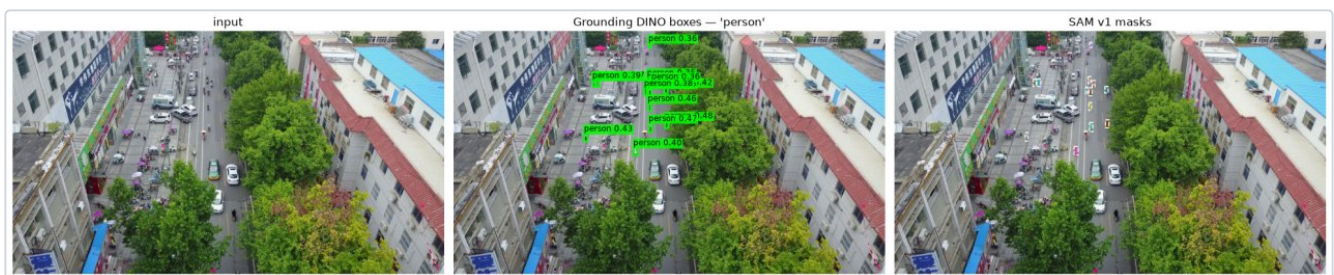


Figure 03. SMALL AERIAL TARGETS. 'person' on a 1920x1080 VisDrone frame -> 13 detections, top score 0.48 (vs 0.67 for 'car' on the same frame). A detector property, identical to the SAM 2 evaluation.



Figure 04. PER-FRAME 'TRACKING', FRAME 192. SAM v1 has no memory, so every frame is an independent detect+segment. Here the highest-confidence 'car' is one of the cars in the centre lanes and SAM v1 masks it cleanly.



Figure 05. THE NEXT FRAME, 193 -- THE NO-IDENTITY FAILURE. The top-1 mask has jumped ~746 px to a completely DIFFERENT car near the top-right horizon. Nothing is wrong with the mask; there is simply no object identity linking frame 193 to 192. 'Which car' is re-decided every frame by confidence (27 such hops in 200 frames on this clip). SAM 2's memory is exactly what prevents this; SAM v1 has no memory to prevent it with.



Figure 06. PER-FRAME zebra.mp4, frame 100. Detection rate is 100% (there is always a zebra in view) and the mask is clean, but throughput is 4.3 FPS because Grounding DINO runs on every frame -- vs 32.5 FPS for SAM 2's detect-once-then-propagate on the same clip. 20 top-1 hops between the ~6 zebras present per frame.